Photonic Interconnects for TPU LLM Workloads

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Motivation

Frontier large language models—open-weight Gemma 4 and the proprietary Gemini 3 family—are trained and served on TPU pods whose performance is bounded less by raw matrix-multiply throughput than by the cost of moving activations, KV-cache state, expert tokens, gradients, and shared weights between thousands of accelerators. TPU v5p and the 9216-chip Ironwood pod (4614 FP8 TFLOP/s per chip, 192 GiB HBM, 7.4 TB/s HBM bandwidth, 1.2 TB/s bidirectional ICI) are organised around high-radix electrical inter-chip interconnect for exactly this reason.

Co-packaged photonic interconnect—of which Lightmatter's Passage family (EVK50, EVK100, L200, L200X, M1000) is a public reference—offers lower per-bit energy and, more interestingly, distance-insensitive transmission and hardware multicast. The question is not just whether links get cheaper; it is whether photonics changes how a TPU pod should be *mapped and scheduled*, and if so, at which layer.

Research question. Is photonic interconnect a constant-factor link substitution, or does it change the optimal partitioning, dataflow, tile sizing, and distributed schedule for an LLM workload running on a TPU-class cluster?

Three insights. (1) The per-chip mapping is invariant under photonics: 0/24 mapping signatures change on the modelled TPU cube. (2) Hardware multicast and distance-insensitive transport, not raw bandwidth, are decisive: M1000 still beats a strong 3.5 Tb/s/6 pJ/bit chiplet baseline by 99.85% on shared-state multicast. (3) Photonics is mostly an inference accelerator: 42.43× Gemma 4 26B sparse-MoE inference speedup vs. 3.37× training, because once collectives shrink training is compute-bound.

Underlying Principles

Four-layer modelling stack. The pipeline has four explicit layers, only three of which the per-chip mapper sees. The first is the *workload* (the matrix-multiply einsum with rank sizes per LLM stage). The second is the *architecture* (a cluster-scale memory level above per-chip HBM, the node-array fanout, the unified buffer, the processing-element array, the weight register, and the MAC). The third is the *mapping* (the sequence of storage, spatial, temporal, and compute directives with concrete tile shapes that the mapper produces— the standard mapping formalism: partitioning, dataflow, tile sizing). The fourth is the *distributed schedule* that lives above the mapper: which tensor a compiler keeps in the cluster pool versus per-chip HBM, and which named XLA-style policy (sharded collective, remote pooled-HBM streaming, replicated cache, photonic multicast cache fill) is selected.

Distance-aware link energy. A bit transferred over h hops of average physical link length $\ell(N)$ pays $E_{\text{bit}}(N) = e_c + h\,\ell(N)\,e_{tx/\text{mm}}$, where e_c is conversion overhead and $e_{tx/\text{mm}}$ is per-millimetre transmission energy. Electrical links pay most of their cost in $e_{tx/\text{mm}}$; photonic links pay it almost entirely in e_c with $e_{tx/\text{mm}} \approx 0$.

Topology cost. $T_{\text{net}}(N) = B/(BW_\ell/C(N)) + h\,t_r + h\,\ell(N)\,t_p + t_s$, with contention $C(N)$ equal to $\sqrt{N}$ for a 2-D mesh, $\sqrt[3]{N}$ for a 3-D torus, and one for a switched fabric. Multicast: a non-multicast technology pays the unicast volume N times for an N-way fanout; a multicast-capable photonic technology pays the unique payload once.

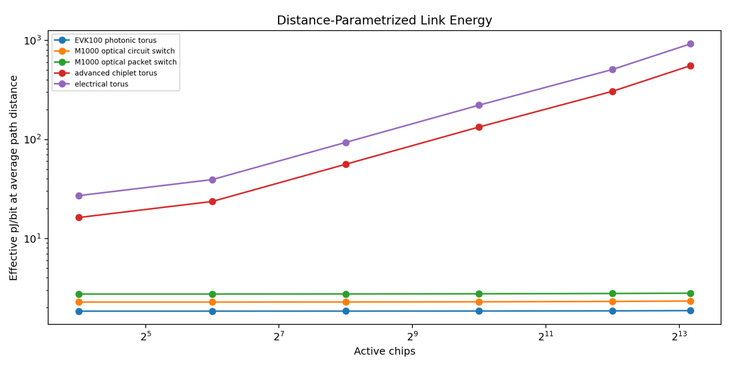
Iterative feedback into the mapper. Run the mapper with the cluster pool exposed as a memory level; price the resulting traffic via the topology cost; fold the per-bit energy and latency back into that level; re-run. Converges in ≤ 3 iterations.

End-to-end stage model. $T_{\text{stage}} = \max(T_c, T_{\text{HBM}}) + (1-\alpha)T_{\text{net}}$ across prefill, decode KV attention, weight or expert refill, MoE dispatch and combine, training forward and backward, gradient synchronization, and optimizer traffic.

5. Evaluation

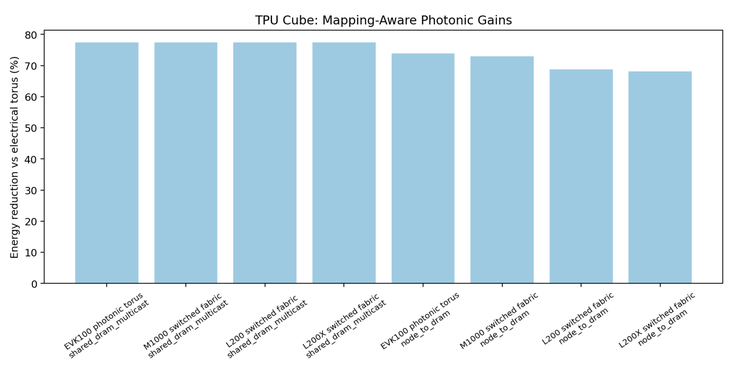
Held fixed. Compute, register, on-chip buffer, per-chip HBM, and HBM energy and latency are identical across all comparisons. Only interconnect technology, topology, and distributed schedule vary. **Modelled technologies.** Electrical 800G (0.8 Tb/s, 10.0 pJ/bit), advanced chiplet (3.5 Tb/s, 6.0 pJ/bit; included specifically to defeat the bandwidth hypothesis), and the Lightmatter Passage points: EVK50 (0.8 Tb/s, 4.6 pJ/bit), EVK100 (3.2 Tb/s, 1.9–3.2 pJ/bit), L200 (16 Tb/s, 4.5 pJ/bit), L200X (32 Tb/s, 4.8 pJ/bit), and M1000 (57.3 Tb/s, 2.3 pJ/bit; built-in optical circuit switching); plus an optical-packet-switching sensitivity at the M1000 bandwidth class. **Workloads.** Six GEMM proxies (decode MLP, KV attention, prefill projection, long-context prefill, MoE dispatch, training weight gradient) drive the per-chip mapper experiments. The end-to-end study uses public Gemma 4 26B sparse-MoE and 31B dense anchors plus Gemini-class Pro MoE and Deep Think long-context stress proxies (parameter counts not public; explicitly labelled). **Reproducibility.** A single command path rebuilds every artifact—figures, CSV outputs, this poster, the report, the technical summary, and the regression suite—inside the course's container image. The regression suite asserts technology constants, the long-distance electrical-vs-photonic crossover, the multicast volume semantics, the per-chip mapping invariance, the strong/weak modes, and the optical-switching sensitivity. **Limitations.** Optical multicast is modelled as one logical broadcast (optimistic, explicit); the stage model is analytical, not cycle-accurate; the OPS point is a sensitivity what-if, not a Lightmatter product claim.

Distance physics: where the crossover lives



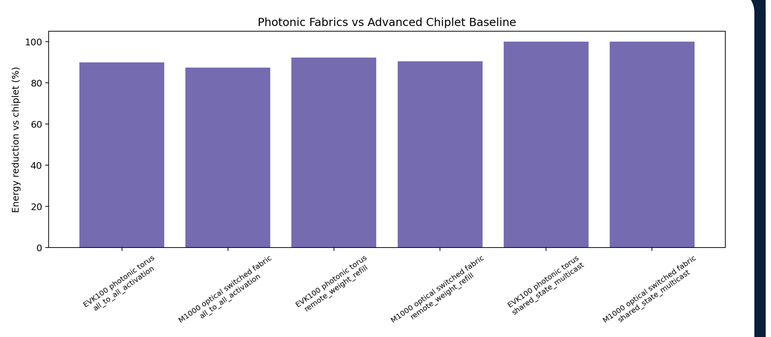
Effective pJ/bit grows nearly linearly for the electrical and chiplet tori as cluster size grows, but stays flat for the photonic switched fabrics. Below ∼10 mm the electrical link wins on raw pJ/bit; near pod-scale path lengths it is two orders of magnitude more expensive per bit. This crossover is the physical reason photonic interconnect cannot be evaluated as a flat constant.

Multicast, not bandwidth, is decisive



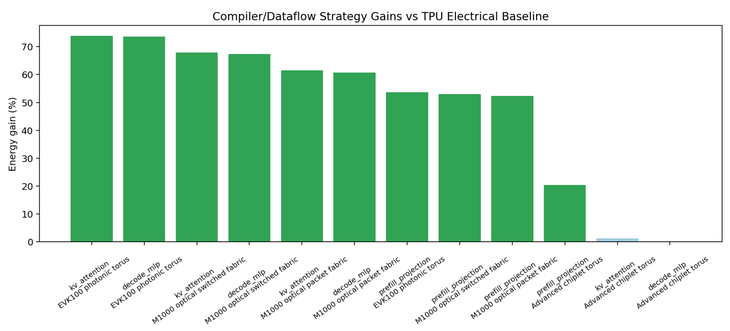
TPU-cube energy reduction relative to the electrical torus, by communication scenario. Multicast bars (left) cluster near 77%; node-to-DRAM bars (right) are smaller and depend more on raw bandwidth. Photonic gain on shared-DRAM multicast reaches 99.93% E / 99.61% L (EVK100) and 99.91% E / 99.99% L (M1000) versus the TPU v5p-class electrical torus.

Bandwidth hypothesis ruled out



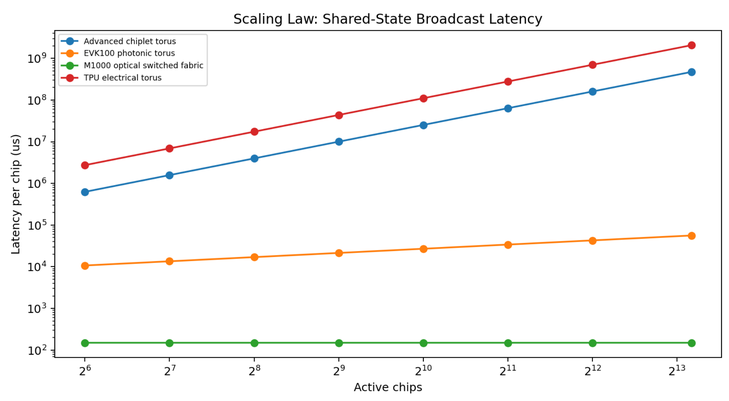
Photonic fabrics versus the strong 3.5 Tb/s, 6 pJ/bit chiplet baseline. The chiplet closes most of the bandwidth gap on activation exchange and remote-weight refill (87–92%), but on shared-state multicast both EVK100 and M1000 still beat it by **99.85%**. N unicasts collapse to one logical broadcast and the contention floor disappears on a switched fabric: multicast is the architectural property that matters.

Layer 3 invariant; layer 4 moves



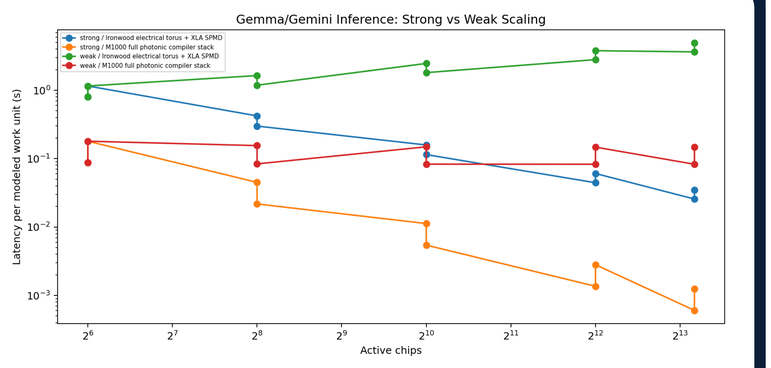
Energy gain of the best XLA-style schedule per (workload, technology) cell relative to the electrical baseline. **0/24** per-chip mapping signatures change under photonics; **5/16** distributed mappings flip from node-to-node to shared-DRAM multicast; remote pooled-HBM streaming on EVK100 removes **73.86%** of the compute energy on KV attention.

Pod-scale scaling: per-chip cost



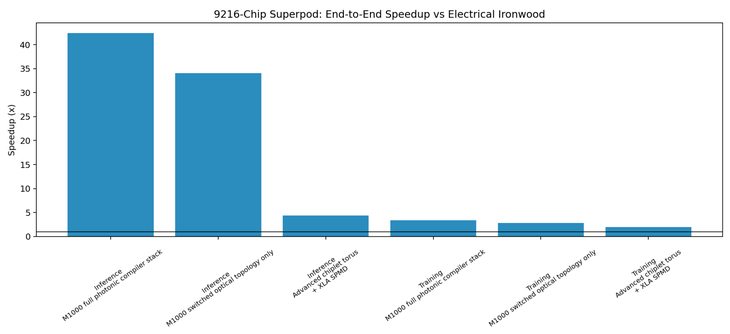
Per-chip shared-state broadcast latency from cube to pod scale on a log–log axis. The TPU electrical torus and the chiplet torus rise nearly two orders of magnitude per decade of chip count; the M1000 switched fabric stays flat. At 9216 chips the gap is roughly seven orders of magnitude.

Strong vs. weak scaling on Gemma 4



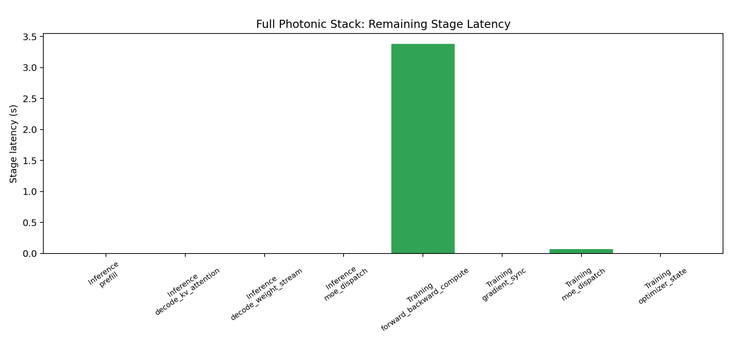
Per-modelled-work-unit latency on Gemma 4 26B sparse-MoE inference under strong (fixed problem on more chips) and weak (per-chip work fixed; problem grows with N) scaling. The full photonic stack maintains a clear advantage at every modelled chip count; weak scaling exposes the per-chip cost growth on the electrical baseline.

End-to-end: inference vs. training



9216-chip Ironwood-pod end-to-end speedup. The Gemma 4 26B sparse-MoE inference proxy reaches **42.43×** latency speedup with the full M1000 photonic stack and 94.05% energy reduction; the larger Deep Think proxy reaches 14.50×. Training reaches only 3.37× on Gemma 4 MoE and 1.43× on the larger proxy.

Why the asymmetry: stage breakdown



Per-stage latency that *remains* once the full M1000 photonic stack has been applied. All inference stages (prefill, decode KV attention, weight stream, MoE dispatch) collapse to near zero. The single stage that does not is training forward/backward compute, which sits inside the $\max(T_c, T_{\text{HBM}})$ envelope and has nothing to do with the network.

Analysis: where it helps, where it does not

Photonics helps most when the workload either fans out to many chips or repeatedly crosses a long path: long-context inference, sparse-MoE serving, remote pooled-HBM access, and weight or KV broadcast. It helps least when the bottleneck is local compute or per-chip HBM, when one nearest-neighbour electrical hop already serves the workload, or when the per-chip mapping was already optimal under the electrical baseline. The per-chip mapping does not change under any photonic technology: that is the empirical justification for separating the per-chip mapping from the cluster-level schedule when reporting any new-technology study. The strong end-to-end case for photonic interconnect at pod scale is therefore inference and long-context serving rather than training payoff. **What worked.** The iterative feedback loop converged in three iterations or fewer in every case; the topology cost model produced the predicted multicast asymmetry; the stage model reproduced the inference–training divergence cleanly. **What did not.** An attempt to fold the inter-chip network into the accelerator architecture as a degenerate memory level produced opaque numbers and was abandoned in favour of the external network model with explicit feedback into a single logical-memory level. **Future work.** Direct integration with an XLA-style compiler so the schedule choice is searched continuously rather than enumerated; a cycle-accurate optical-switch simulator; co-design of MoE expert placement under photonic locality; and pipeline-parallel and ZeRO-3 training stage models.

References

[1] V. Sze and J. Emer, MIT 6.5930 lecture notes: *Memory & Metrics*, *Mapping*, *Mapping/Partitioning*, *Advanced Technologies*, Spring 2026. [2] M. Gilbert, T. Andrulis, V. Sze, J. S. Emer, "Turbo-Charged Mapper," MIT, 2025. [3] T. Andrulis et al., "Architecture-Level Modeling of Photonic DNN Accelerators," ISPASS 2024. [4] Lightmatter, Passage product family / EVK100 / M1000 EVK / L200 product pages and SC25 & ISES 2025 talks (`lightmatter.co`). [5] Google Cloud, "TPU v5p," "Cloud TPU7x" (Apr. 2026), and the Ironwood blog; P. Kanwar, "TPU Rack Scale Systems," Hot Chips 2025 Tutorial. [6] PyTorch/XLA, "SPMD user guide." [7] Google, "Gemma 4 open-model release," 2026.

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